Modeling of kilonova light curves from neutron star mergers and possible constraints on nuclear physics

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
in partial fulfillment of the requirements for the degree of

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ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. When two neutron stars merge, part of their mass is ejected and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, this ejected material is a site of rapid neutron capture (r-process) nucleosynthesis. The yield of heavy nuclei in the ejecta depends on the properties of neutron-rich nuclei far from stability, including nuclear masses, decay rates, neutron-capture rates, and fission properties, among others, many of which are unknown. Since nuclear physics plays a crucial role in the evolution of observables, understanding the impact of relying on theoretical nuclear physics results is essential to correctly interpret current and future multimessenger observations.

This work presents a detailed analysis of the impact of different theoretical α- and β^--decay rates on the kilonova evolution. The nucleosynthesis was calculated using the nuclear reaction network WinNet across an ensemble of initial thermodynamic conditions parametrized by the initial electron fraction ($Y_{e,0}$), the expansion timescale ($\tau_{\rm exp}$) and the initial entropy per baryon (s_0). The kilonova light curves were calculated using current semi-analytical models based on radiative-transfer simulations. The results show that the heating rate from radioactive decays deviates from the widely accepted power-law behavior depending on the β^--decay prescription consistently across the initial conditions studied. In addition, the set of nuclei responsible for such behavior was identified and the description of their decay chains is presented.

The results on the impact of theoretical rates described above are then tested in a merger simulation where the resulting kilonova is now described as a superposition of different ejecta components[1]. The amount of ejected mass and its velocity are extracted from a general-relativistic simulation which was computed using the Einstein Toolkit framework.

Thesis supervisor: Neelima Govind Kelkar
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[1]This paragraph may change as results from GR simulations are being obtained

Acknowledgments

Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

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Chapter 1

Introduction

Binary neutron star mergers (BNSMs) are among the most energetic events in the Universe, releasing tremendous amounts of energy through gravitational waves, neutrinos, and electromagnetic radiation. During and after the merger of two neutron stars, a fraction of their mass is ejected through tidal interactions, shocks, and disk winds. The ejected material is expected to consist of extremely neutron-rich matter, within which rapid neutron captures (r-process) lead to the formation of heavy nuclei far from stability. Consequently, these mergers are considered one of the primary sites for the synthesis of the heaviest elements in the Universe [1, 2]. As the freshly synthesized nuclei decay toward stability, their radioactive energy release drives the thermodynamic evolution of the ejecta and produces a thermal electromagnetic transient [3]. Such a transient is called a kilonova and will be the main topic of this thesis, since it encodes crucial information about the physics of the merger, such as the abundance pattern of the elements and the amount of ejected material.

1.1 Kilonovae in a nutshell

Kilonovae are closely related to the nuclear and atomic phenomena inside the ejecta. As heavy nuclei decay toward stability, part of the energy is released in the form of high energy photons according to the specific transitions inside the nuclei. In addition, decay products such as α and β particles, which also carry part of the decay energy, interact with surrounding charged particles primarily through Coulomb interaction, producing photons that carry away part of that energy. Effectively, the relevant particles will be photons.

At early stages the density is still high, so photons cannot easily escape and interact with several charged particles before reaching the surface. As photons interact with matter, their energy spectrum approaches that of a blackbody in a process called thermalization, and the same process applies to all the other decay products [4]. The emission at such stages is well described as a blackbody with a temperature evolution governed by the heating rate from radioactive decays. As the density decreases, bound states such as atoms form in a process called recombination. After this point the thermalization of the photons depends on the emission and absorption spectra of heavy elements. As a consequence, kilonova light curves encapsulate information about the abundance of certain nuclei with specific emission and absorption lines. For example, the presence of lanthanides increases the opacity of the ejected

material by an order of magnitude, so a kilonova with lanthanides will thermalize more efficiently and develop a long-lived thermal spectrum [5]. Since the abundances of lanthanides and of the rest of nuclei are determined by nuclear physics inputs, such as reaction rates, decay rates and masses, a detailed treatment of their uncertainties is needed for the correct interpretation of current and future observations. This is where this research work sits, as developed later in the text.

In the following sections, we will first give the reader a brief historical introduction to the kilonova. Then we will present modern kilonova models and their limitations. By the end of this chapter, we will state the objective of the research, explain the methodology, and describe the structure of this thesis.

1.2 A brief history of the kilonova

Until the last decade, the astrophysical origin of roughly half of the nuclei heavier than iron remained one of the major open problems of nuclear astrophysics. Big Bang nucleosynthesis accounts for the primordial abundances of the lightest nuclides (D, ^{3}He, ^{4}He and ^{7}Li), while hydrostatic and explosive burning in stars and supernovae produces the elements from carbon up to the iron peak and some heavier elements [6, 7]. The issue was that, beyond iron, the binding energy per nucleon decreases and Coulomb barriers grow, so heavy nuclei are built mainly by neutron captures rather than by charged particle fusion. Two main channels were known [8]. The first channel is the slow (s) process, in which neutron captures are slower than β-decays. The second is the rapid (r) process, in which captures are much faster than β-decays and drive matter far to the neutron-rich side of stability. The s-process is now well understood and operates mainly in asymptotic giant branch (AGB) stars and, for the lighter heavy elements, in massive stars [9]. The conditions required by the r-process have also been characterized. Those are neutron densities $n_n \gtrsim 10^{20}\,\mathrm{cm}^{-3}$, or equivalently suitable combinations of electron fraction, entropy and expansion timescale in the ejecta [10]. However, the site where these conditions occur remained elusive.

For several decades the favored candidate was the core-collapse supernova, in particular the neutrino-driven wind from the newborn neutron star [11]. Modern simulations, however, find this wind neither neutron-rich nor entropic enough to produce the heaviest r-process nuclei [12], so ordinary supernovae contribute at most a weak r-process, producing elements up to about Sr–Ag. Observations point in the same direction. The large star-to-star scatter of europium in metal-poor stars [13], the strong r-process enrichment of the ultra-faint dwarf galaxy Reticulum II [14], and the low abundance of live ^{244}Pu in deep-sea sediments [15] all indicate that the heavy r-process elements are made in rare events with large yields, rather than in frequent ordinary supernovae.

Neutron star mergers, on the other hand, match those requirements. Early theoretical studies had proposed that during the merger of compact objects, tidal forces lead to the ejection of neutron-rich material [16]. These ejecta were expected to be very neutron-rich and decompress rapidly, ideal for the r-process, enabling the synthesis of heavy neutron-rich nuclei. It was then proposed that the radioactive decay of freshly synthesized nuclei would power a short-lived optical transient. In 1998, Li and Paczyński presented a simple model (LP98) to calculate that possible electromagnetic counterpart. The model predicted a light

curve with a bolometric luminosity peak of approximately 10^{44} erg s^{-1} at $t \approx 1$ day after the merger depending on free parameters such as the expansion velocity, the ejected mass and an effective constant for the radioactive heating [17].

1.3 Modern models of kilonovae and observations

The LP98 model assumes an analytical energy generation rate of the form $\dot{\varepsilon}(t) \propto t^{-1}$ and models the electromagnetic emission as a blackbody spectrum. However, these assumptions may not fully capture the physical complexity of the event. As discussed before, the γ-rays produced by nuclear decays can escape without thermalizing, leading to deviations in the predicted emission spectrum. In addition, the actual energy generation rate may deviate from the assumed analytical form, making a more detailed treatment of the spectral evolution of the emission a challenging task.

The first attempt to construct a realistic model of this phenomenon was presented by Metzger et al. [3], who performed self-consistent calculations of the optical and near-infrared counterparts with two major improvements. First, they implemented nuclear reaction network calculations to evolve the abundances and compute the corresponding energy generation rate, $\dot{\varepsilon}(t)$, using both experimental and theoretical reaction and decay rates. Second, they performed one-dimensional Monte Carlo radiative transfer simulations, assuming opacities derived from bound–bound transitions of iron (Fe), with corrections applied to the ionization energies, in order to model the resulting light curves. The first results showed an abundance pattern close to the solar system r-process abundances. Also, the energy generation rate was found to follow a power-law dependence, $\dot{\varepsilon}(t) \propto t^{-\alpha}$ with $\alpha \sim 1.1$–1.3, and remarkably robust against variations in the electron fraction ($Y_e \sim 0.05$–0.35), thermodynamic trajectories, and nuclear properties along the r-process path. The second key result demonstrated that the light curves evolve significantly differently from those predicted by the simple blackbody approximation. The simulated light curves exhibited a characteristic "bump-like" behavior on different timescales across various wavelength bands and a peak luminosity three orders of magnitude greater than that of a typical nova, the latter being the reason why the term kilonova was first introduced.

However, radiative transfer simulations face major challenges due to the highly complex opacities arising from the dense line spectra of heavy r-process nuclei. This "opacity problem" has been addressed through several approaches. For instance, in Ref. [18], atomic structure models for representative elements were employed to compute radiative transition rates for tens of millions of lines, enabling more realistic estimates of the opacities for r-process material. A different approach was taken in Ref. [19] where the opacity was estimated by taking into account electron scattering, free–free, bound–free and bound–bound transitions using analytical formulas. These advances led to the identification of distinct kilonova components based on the lanthanide content of the ejecta [20, 21]. Material with a low electron fraction ($Y_e \leq 0.22$) is expected to produce lanthanide-rich ejecta with high opacity, resulting in a red, longer-lived transient commonly referred to as the red kilonova. In contrast, material ejected with a higher electron fraction ($Y_e \geq 0.3$), typically from the polar regions, evolves into lanthanide-poor ejecta with lower opacity, producing a short-lived, blue optical transient known as the blue kilonova.

Nevertheless, all previous efforts remained purely theoretical and computational, as no observational data were available to validate the models. This situation changed on August 17, 2017, when the Advanced LIGO and Advanced Virgo gravitational-wave detectors made their first observation of a binary neutron star merger (BNSM) [22] with the subsequent identification of the electromagnetic counterparts in the same location, which supported the identification of the event as a BNSM. An approximately 100 s long gravitational-wave signal, GW170817, was followed by a short gamma ray burst (sGRB), GRB 170817A, and an optical transient named SSS17a/AT 2017gfo [23], found in the host galaxy NGC 4993. The source of the signal was detected across the electromagnetic spectrum in the X-ray, ultraviolet, optical, infrared, and radio bands over hours, days, and weeks [24].

The electromagnetic transient SSS17a/AT 2017gfo was of great significance, as it provided the first compelling observational evidence for the nucleosynthesis of heavy nuclei through the r-process. As shown in Ref. [25], the observed emission was well reproduced by a three-component kilonova model: a lanthanide-poor blue component with an opacity of $\kappa = 0.5$ $\mathrm{cm^2\,g^{-1}}$, ejecta mass $M_{\rm ej} \approx 0.020\,M_\odot$ and velocity $v_{\rm ej} = 0.27c$; an intermediate (purple) component with $\kappa = 3$ $\mathrm{cm^2\,g^{-1}}$, $M_{\rm ej} \approx 0.047\,M_\odot$ and $v_{\rm ej} = 0.15c$; and a lanthanide-rich red component with $\kappa = 10$ $\mathrm{cm^2\,g^{-1}}$, $M_{\rm ej} \approx 0.011\,M_\odot$ and $v_{\rm ej} = 0.14c$. Meanwhile, numerous efforts have been made to improve previous kilonova models, motivated by the prospect of obtaining high-resolution data from future observations with the James Webb Space Telescope (JWST). However, performing fully consistent general relativistic simulations that include a realistic treatment of nuclear matter with proper neutrino transport, followed by detailed nucleosynthesis calculations in the ejecta and subsequent radiative transfer simulations to reproduce the observed light curves, remains computationally demanding. Consequently, several practical semi-analytical formulations have been proposed to approximate key physical processes and reduce the computational cost while retaining the essential physics of the problem.

One of the cornerstones of analytical modeling was presented in Ref. [26], which introduced a simplified model not intended to replace full multi-group radiative transfer simulations, but rather to serve as a toy model for qualitatively understanding the evolution of the transient. More refined analytical approaches were proposed shortly thereafter. For instance, Ref. [27] developed a model capable of reproducing multi-band light curves by assuming a time-dependent effective opacity and a uniform power-law energy generation rate. Some models also account for the impact of ejecta morphology on the observed emission, introducing more realistic prescriptions for non-spherical geometries and their influence on the angular dependence of the kilonova light curves [28].

The present panorama of kilonova modeling extends beyond purely analytical formulations and is now better represented by semi-analytical frameworks, such as the one introduced in Ref. [29]. This framework provides a practical approach for generating broadband light curves that account for multiple ejecta components and non-spherical geometries, while incorporating time- and composition-dependent heating rates, thermalization efficiencies, and opacities.

Such advances in kilonova modeling have opened new possibilities for studying distinct features in the light curves that may contain additional information about the composition of the ejecta. It has been theorized that the late time kilonova emission, occurring weeks to months after the merger, can reveal the radioactive signatures of specific heavy nuclei, rather

than the power-law heating that dominates the early time emission [30]. Isotopes such as ^{223}Ra ($t_{1/2} = 11.43$ d) and ^{225}Ac ($t_{1/2} = 10.0$ d), which undergo α-decay, and ^{254}Cf, which undergoes spontaneous fission, have been identified as potential dominant contributors to the late-time heating [31].

These nuclei release large amounts of energy per decay and the α particles thermalize more efficiently than typical β-decay products. Therefore, these nuclei are expected to produce detectable bumps or departures from a smooth power-law decline in the bolometric light curve. However, identifying such "fingerprints" requires sufficiently accurate heating-rate predictions from nucleosynthesis calculations and sensitive, multi-epoch, wide-band observations.

In March 2023, the electromagnetic transient AT 2023vfi, associated with the gamma-ray burst GRB 230307A, was identified as the second confirmed observation of a kilonova [32]. This signal has provided an early opportunity to apply these ideas observationally [33]. JWST follow-up analyses indicate that AT 2023vfi's early evolution resembles AT 2017gfo during the first week, and detailed spectral/photometric modeling has been used to place constraints on the presence and abundances of heavy nuclei in the ejecta. The work presented in Ref. [34] reports upper limits and tentative abundance estimates for isotopes such as ^{222}Rn, ^{223}Ra, ^{224}Ra, and ^{225}Ac. These results illustrate both the promise and the current limitations: while JWST data can probe late-time radioactive contributors, robust identification of specific isotopes requires careful propagation of nuclear-physics (decay-rate and fission-yield) and opacity uncertainties. In this context, one of the goals of this work is to assess whether a more accurate treatment of decay-rate uncertainties can improve the constraints on the abundance of heavy nuclei.

1.4 Objectives and thesis structure

As mentioned before, the main objective of this work is to evaluate the sensitivity of kilonova light curves to the choice of theoretical decay rates prescription. We constructed an ensemble of possible initial thermodynamic trajectories within the ejecta. For each initial condition, we ran six nuclear reaction network calculations with three different β^--decay and two α-decay prescriptions. Then, with the results for the abundances and the heating rate, we modeled the kilonova. We found consistent deviations from the power-law behavior across the ensemble of initial conditions. We also identified the nuclei responsible for such behavior along with their decay chains.

The thesis is divided into three main parts. The introductory part consist of Chapters 1 and 2. Chapter 2 develops the physics needed to understand neutron stars and binary neutron stars. Chapters 3, and 4 present the basic theoretical framework from literature for the stages of the merger simulation: the nuclear reaction network evolution and the construction of the kilonova light curve from emission models. The last part, Chapters 5, 6, 7and 8, will explain the details of how this work constructed the initial thermodynamic trajectories, the setup we used for the nuclear reaction network calculations with their results and the construction of the kilonova light curves.

Chapter 2

From Neutron Stars to Kilonovae

Kilonovae are the main topic of study throughout this thesis. Every chapter and section was built with the objective of developing the intuition on all the processes that take place in kilonovae. However, the necessary astrophysical context will enrich our understanding. Therefore, this chapter will first present the neutron star and its dense matter in the general relativity description. Then the text will move to the problem of finding two neutron stars in a close enough orbit and present a brief discussion of why the existence of such configurations is not trivial. After that, a description of the different phases of the merger will be presented. The different emission mechanisms will be discussed: gravitational waves, short gamma ray bursts and our favorite the kilonova.

2.1 The Neutron Star

Neutron stars contain the densest matter known to exist in a stable state. Their central densities reach several times the nuclear saturation density $n_{\rm sat} \simeq 0.15$ fm^{-3} [35]. In the outer layers of the star, nuclei coexist with a degenerate electron gas. With increasing density, electron captures make the nuclei progressively more neutron-rich until, at the neutron-drip density $n_{\rm drip} \simeq 2.5 \times 10^{-4}$ fm^{-3} ($\rho \simeq 4 \times 10^{11}$ g cm^{-3}), neutrons are no longer bound to the nuclei and form a free neutron gas [36]. At about half the saturation density, the distinction between the inside and the outside of a nucleus effectively vanishes and matter becomes a continuum of nucleons, which marks the transition from crust to core. At the same time, the β equilibrium favors neutrons instead of protons, which leads to a neutron dominated composition.

Observed neutron-star masses range from about 1.1 to slightly above $2\,M_\odot$, and their radii are inferred to lie between roughly 10 and 14 km [37, 38]. However, a free degenerate neutron gas can support stars of at most $\simeq 0.7\,M_\odot$ [39, 40]. To explain the observations accurately, the strongly repulsive short-range part of the nuclear interaction is required. The equation of state (EOS), therefore, has to include nuclear interaction terms, which is a formidable task to calculate.

The microscopic theory of this matter, quantum chromodynamics (QCD), cannot at present provide the EOS in the relevant density range. At the baryon chemical potentials of neutron-star cores, QCD is non-perturbative, and lattice QCD is hampered by the fermion

sign problem at finite baryon density. Controlled calculations exist only in two limits: chiral effective field theory describes nuclear matter up to about 1–2 $n_{\rm sat}$ [41], and perturbative QCD becomes reliable only at densities of order 40 $n_{\rm sat}$, far above those realized in neutron stars [42]. In the intermediate regime, where neutron-star cores reside, the EOS is described by phenomenological models. As a consequence, a first principles description of neutron star cores is still lacking. Matter at several times $n_{\rm sat}$ may consist of nucleons, may contain hyperons, or may undergo a transition to deconfined quark matter, either through a smooth crossover or through a first order hadron-quark phase transition [43].

Let us assume, for instance, that we know the EOS. We can then estimate the radius of the star for a certain range of masses, including GR effects, through the Tolman-Oppenheimer-Volkoff equation:

$$\frac{dP}{dr} = -\frac{G}{r^2}\left(\rho + \frac{P}{c^2}\right)\left(m + 4\pi r^3 \frac{P}{c^2}\right)\left(1 - \frac{2Gm}{c^2 r}\right)^{-1}. \tag{2.1}$$

The equation assumes a static, spherically symmetric perfect fluid. This is a good approximation, but real stars have angular momentum, so rotational corrections will be relevant. Here $\rho(r)$ and $P(r)$ are the density and pressure at a radial distance r. The quantity $m(r)$ is the mass within a sphere of radius r. At this point we have not introduced any EOS yet. The relationship between P and ρ could depend on temperature and composition. Many current EOS models actually present their results for a range of temperatures and electron fractions Y_e. Once the EOS is established, one can find a mass–radius curve. By comparing with observations one can in principle infer constraints on the EOS. Such studies have been done, for example the existence of pulsars with masses of about 2 $M_\odot$ [44] and the radii inferred by NICER [38] constrain the pressure of dense matter up to densities of roughly $2-3\,n_{\rm sat}$ [45]. These measurements, however, constrain the relation between pressure and energy density rather than giving a full description of the matter. For example, hybrid stars (neutron stars with quark cores) can have mass–radius relations nearly indistinguishable from those of purely hadronic stars [46].

2.2 Formation of binary neutron stars

For two neutron stars to merge, three conditions must be met. Both stars must form, which requires two core-collapse supernovae (SNe), and the binary must survive both explosions. Then the two remnants must end up close enough for gravitational-wave (GW) emission to bring them into contact in less than the age of the universe. For typical neutron-star masses, the initial separation must be smaller than about 5 $R_\odot$. This is two to three orders of magnitude smaller than the radii that the progenitor stars reach as supergiants. The central problem of binary-neutron-star formation is therefore how to bring the compact remnants of two massive stars to separations far smaller than the stars themselves once were, without unbinding the system on the way.

In the standard isolated channel, a double neutron star (DNS) descends from a binary of two OB stars that are massive enough to end their lives in a core-collapse SN (see Fig. 2.1). If the stars of the initial binary are too far apart, the resulting DNS will not be close enough to merge. If, on the contrary, the initial stars are close enough, they will interact via stable

or unstable mass transfer. Once one of the stars reaches the end of its life, it expands and transfers part of its mass to its companion. This star then explodes in a SN and leaves a NS, as in Fig. 2.1. If the two stars remain bound after the supernova, they become a high-mass X-ray binary (HMXB). When the second star expands, it engulfs the first star in a phase called common envelope [47]. In that stage, the dynamical friction on the first star as it moves inside the envelope of the second one causes the loss of angular momentum needed to bring the cores closer. This phase is not stable, so eventually the envelope either falls onto the stars or is ejected, leaving behind the first NS and the He core of the second star, which eventually dies in what is called an ultra-stripped SN [48]. Finally, the system results in a DNS close enough to merge. The sequence of events is reviewed in [49].

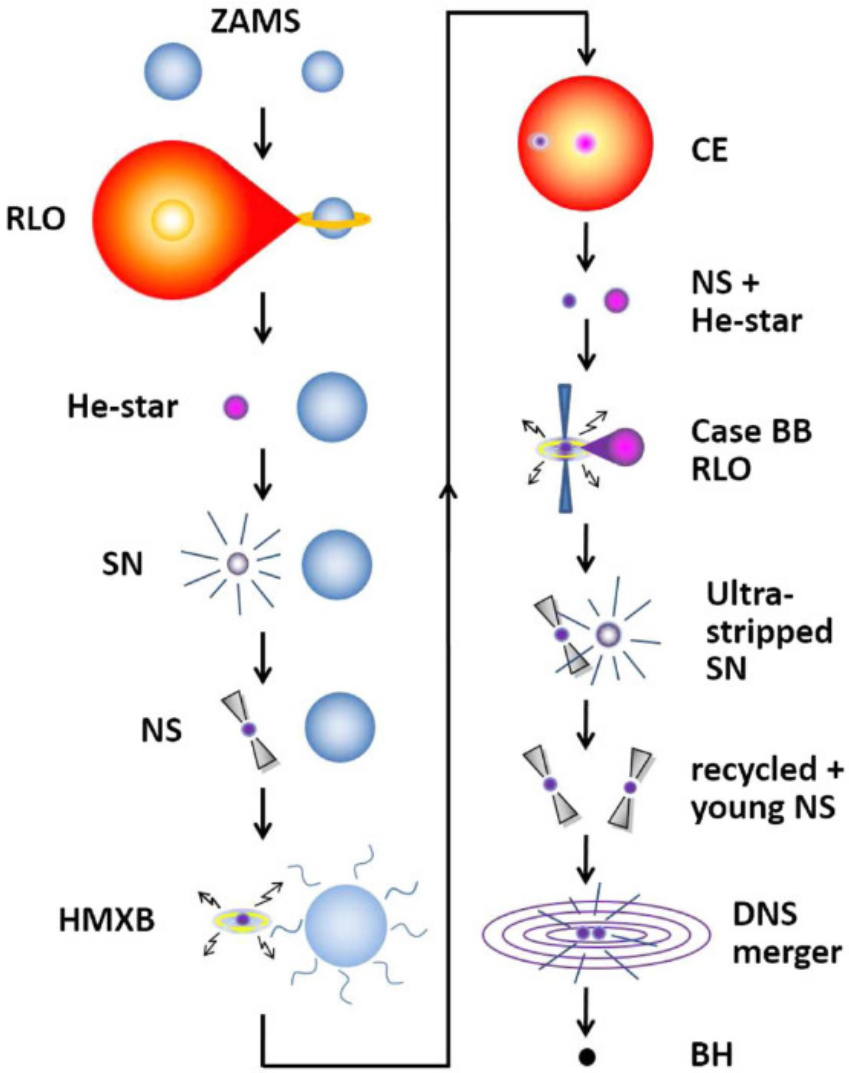


Figure 2.1: Double neutron star (DNS) formation from two OB stars as presented in [49]. ZAMS refers to zero-age main sequence, RLO Roche-lobe overflow, SN supernova, HMXB high-mass X-ray binary and CE common envelope.

2.3 Gravitational-wave-driven inspiral and the maximum separation

Once both neutron stars have formed, the only process that changes the orbit appreciably is the emission of gravitational radiation. Here we will show an estimate of the time to merger at leading (quadrupole) order and the maximum initial separation that allows a merger within the age of the Universe. This calculation is based on the work presented in [50].

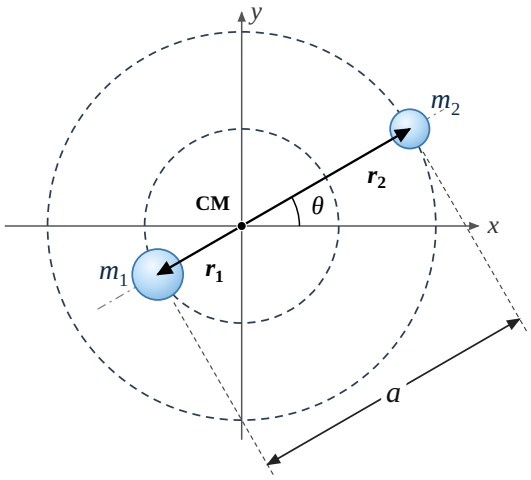


Figure 2.2: Two neutron stars with masses m_1 and m_2 with separation a seen from the center of mass frame.

Consider two point masses m_1 and m_2 on a circular orbit of separation a and angular frequency $\omega = \sqrt{GM/a^3}$, with total mass $M = m_1 + m_2$ and reduced mass $\mu = m_1 m_2/M$. Their position vectors are $\mathbf{r}_1$ and $\mathbf{r}_2$, respectively (see Fig. 2.2). Define the quadrupole mass tensor as

$$Q_{ij} = \sum_l m_l r_{li} r_{lj}, \tag{2.2}$$

summing over the masses m_l with $r_{li} = \mathbf{r}_l \cdot \hat{\imath}$. From this configuration, the positions are

$$\mathbf{r}_1 = -\frac{\mu a}{m_1}\hat{n}\,; \quad \mathbf{r}_2 = \frac{\mu a}{m_2}\hat{n} \quad \text{with} \quad \hat{n} = (\cos(\theta), \sin(\theta), 0). \tag{2.3}$$

So the quadrupole mass tensor components are:

$$Q_{xx} = \frac{\mu^2 a^2}{m_1}\cos^2(\theta) + \frac{\mu^2 a^2}{m_2}\cos^2(\theta) = \mu a^2 \cos^2(\theta) = \frac{\mu a^2}{2}(1 + \cos(2\theta)) \tag{2.4}$$

$$Q_{yy} = \mu a^2 \sin^2(\theta) = \frac{\mu a^2}{2}(1 - \cos(2\theta)) \tag{2.5}$$

$$Q_{xy} = \mu a^2 \sin(\theta)\cos(\theta) = \frac{\mu a^2}{2}(\sin(2\theta)) \tag{2.6}$$

Fortunately, the power radiated in the form of GWs ($L_{\rm GW}$) can be estimated using Eq. (6) of Ref. [50].

$$L_{\rm GW} = \frac{G}{5c^5}\left(\dddot{Q}_{ij}\dddot{Q}_{ij} - \frac{1}{3}\dddot{Q}_{ii}\dddot{Q}_{jj}\right) \tag{2.7}$$

$$L_{\rm GW} = \frac{G}{5c^5}\left(\dddot{Q}_{xx}{}^2 + \dddot{Q}_{yy}{}^2 + 2\dddot{Q}_{xy}{}^2 - \frac{1}{3}(\dddot{Q}_{xx} + \dddot{Q}_{yy})^2\right). \tag{2.8}$$

Since $\dddot{Q}_{xx} = -\dddot{Q}_{yy}$ the last term vanishes and with $\dddot{Q}_{xx}^2 + \dddot{Q}_{yy}^2 + 2\dddot{Q}_{xy}^2 = 32\,\mu^2 a^4\omega^6$ (see Appendix A), the GW luminosity is

$$L_{\rm GW} = \frac{32}{5}\frac{G}{c^5}\mu^2 a^4\omega^6 = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2 M}{a^5}. \tag{2.9}$$

The energy radiated is drawn from the orbital energy $E = -Gm_1m_2/(2a)$. Setting $\mathrm{d}E/\mathrm{d}t = -L_{\rm GW}$ gives

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{\beta}{a^3}, \qquad \beta \equiv \frac{64}{5}\frac{G^3 m_1 m_2 M}{c^5}. \tag{2.10}$$

Equation 2.10 integrates to $a^4(t) = a_0^4 - 4\beta t$, so a binary with initial separation a_0 merges ($a \to 0$) after

$$t_m(a_0) = \frac{a_0^4}{4\beta} = \frac{5}{256}\frac{c^5 a_0^4}{G^3 m_1 m_2 M}. \tag{2.11}$$

The steep a_0^4 dependence is the key point. Halving the separation shortens the inspiral by a factor of sixteen.

Now let us estimate the maximum separation for BNS to merge within the age of the Universe. Note that, since the stars require a prior formation and evolution, this calculation overestimates the maximum distances between stars. Let us require $t_{\rm m} \leq t_{\rm U}$, with $t_{\rm U} \simeq 13.8$ Gyr the age of the Universe

$$a_{\rm max} = (4\beta\, t_{\rm U})^{1/4} = \left(\frac{256}{5}\frac{G^3 m_1 m_2 M}{c^5}\, t_{\rm U}\right)^{1/4}. \tag{2.12}$$

For $m_1 = m_2 = 1.35\,M_\odot$, $m_1 m_2 M = 4.92\,M_\odot^3$ and $\beta = 6.08 \times 10^{27}$ cm^4 s^{-1}. Hence

$$a_{\rm max} \simeq 3.2 \times 10^6 \text{ km} \simeq 4.6\,R_\odot, \tag{2.13}$$

about 0.02 AU. As reference, red supergiants have typical radii of hundreds of $R_\odot$, so having neutron stars closer than $4.6\,R_\odot$ is quite a challenge.

2.4 Mass ejection: tidal, shock-driven, and disk-wind ejecta

At the last instants of the inspiral phase, when the separation becomes comparable to a few stellar radii, the point mass approximation is no longer accurate. From those last instants

to the first ones after the merger the system becomes really complex to study. Physical phenomena and processes such as deformation, dense matter compression and heating, neutrino interactions and the appearance of strong magnetic fields become relevant. Even though current merger simulations include specialized treatment for each of the previously mentioned effects there remain loopholes in our understanding [51, 52]. From the point of view of nucleosynthesis, the important outcome of the merger is the matter that becomes gravitationally unbound. This matter is what we will call the ejecta and is divided into two categories according to the mechanism of ejection [26]. The first is the dynamical ejecta, it is expelled within ~ 10 ms of the merger by tidal torques (tidal ejecta) and by shocks at the collision interface (shock-heated ejecta). The second is the secular or post-merger ejecta, it is expelled after the dynamical ejecta by winds from the remnant and from the accretion disk that surrounds it.

2.4.1 Dynamical ejecta

The study of dynamical ejecta is closely related to numerical relativity simulations. Results from such simulations help us to estimate really important properties of the ejected material such as masses, velocities, distribution, temperatures and electron fraction depending on the sophistication of the simulation. Current results from simulations estimate that the amount of dynamical ejecta lies between $\sim 10^{-4}$ and $\sim 10^{-2} M_\odot$, with asymptotic velocities of 0.1–0.3 c [53]. Electron fraction and distribution depend on the exact mechanism. Tidal ejecta usually have low electron fraction ($Y_e < 0.15$) and are ejected mostly through the equator. Shock driven ejecta on the other hand could reach higher electron fractions ($Y_e < 0.4$) and could be emitted at angles close to the poles (see Fig. 2.3). One interesting result is that the channels are not equally likely to dominate. The tidal component dominates for unequal masses and stiff EOS while the shock component dominates for equal masses and soft EOS. In addition, the total ejecta mass correlates with the tidal deformability and with the mass ratio and there are fits calibrated on simulations for such quantities [54].

2.4.2 Post-merger ejecta

Post-merger ejecta is quite interesting since its mechanism relies on the presence of winds from the remnant. So, let us talk a little bit about the remnant. BNSMs have two possible outcomes, they become a black hole or another neutron star. The first outcome happens when there is a collapse and the interesting fact about this is that it could happen at any time during the merger. For example, the binary could become a very heavy unstable neutron star (long-lived remnant) that, after it cools down (typically seconds), collapses to a black hole. The collapse can happen also at the moment of the merger (prompt collapse) or during a viscous phase shortly after (short-lived remnant).

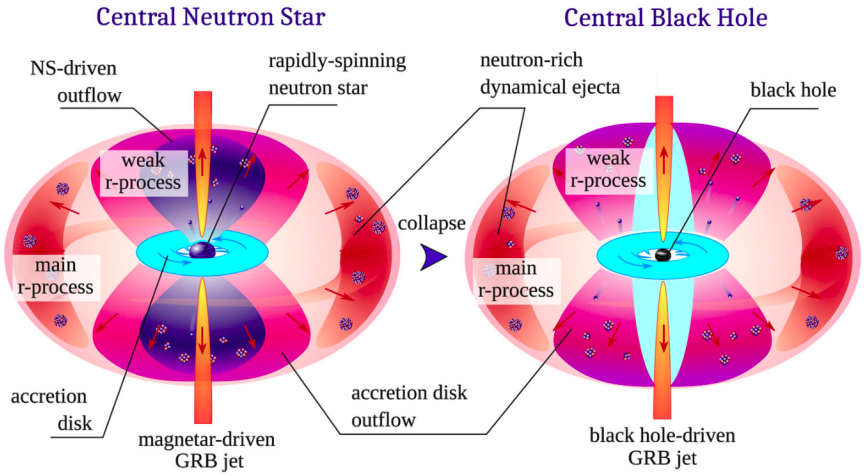


Figure 2.3: Diagram of the remnant from BNSM. Figure taken from Ref. [2].

In all but the most prompt collapses, part of the material with high angular momentum settles into a rotationally supported torus around the central object, with masses $\sim 10^{-3}$–$0.2 M_\odot$. At early times the disk is dense, hot and has low electron fraction. As the disk spreads the viscous heating and the synthesis of α particles and heavier nuclei drive the outflow [55]. As a result part of the disk accumulates enough kinetic energy to unbind from the remnant contributing to the kilonova.

2.5 Weak interactions and the electron fraction of the ejecta

Before the merger, the stars can be described as cold neutron rich matter. At the merger, as we mentioned, the matter heats up and several effects take place. As the matter moves from the β equilibrium, neutrinos become important since they control the Y_e distribution across the ejecta. This is why neutrino transport schemes are important in merger simulations. Simulations without them mispredict the Y_e and are not useful for abundance calculations. This section will establish first why Y_e is important and then how neutrinos control it.

The outcome of the r-process in an expanding fluid element is set mainly by three quantities evaluated when nuclear statistical equilibrium freezes out: the electron fraction Y_e, the entropy per baryon s, and the expansion timescale $\tau_{\rm exp}$ [21]. They fix the neutron-to-seed ratio at the onset of neutron captures. Merger ejecta have moderate entropies, $s \sim 10$–$30\, k_B$ per baryon, and short expansion times, $\tau_{\rm exp} \sim$ ms, so Y_e is the dominant parameter. By

charge neutrality,

$$Y_e = \frac{n_{e^-} - n_{e^+}}{n_b} = \sum_i Z_i Y_i, \qquad Y_i = \frac{X_i}{A_i}, \tag{2.14}$$

where X_i is the mass fraction (ρ_i/ρ), A_i and Z_i are the mass number and proton number of species i respectively and $n_{e^\pm}, n_b$ are number densities of positrons, electrons and baryons. Strong and electromagnetic reactions conserve Y_e. It changes only through charged-current weak interactions.

In a cold neutron star the neutrinos escape freely ($\mu_\nu = 0$) and matter is in neutrinoless β equilibrium, $\mu_n = \mu_p + \mu_e$[1]. Because μ_e grows rapidly with density, equilibrium favors neutrons. In the inner crust and outer core, from which the dynamical ejecta originate, Y_e has values just above neutron drip to ≈ 0.03–0.1 at $\rho \sim 10^{13}$–10^{14} g cm^{-3}. If this matter were ejected without further weak processing, it would have a neutron-to-seed ratio of several hundred and produce a robust abundance pattern from the second peak up to the actinides [56]. During and after the merger, however, the ejecta are heated and irradiated, and two classes of reactions can change Y_e:

$$e^- + p \rightleftarrows n + \nu_e, \qquad e^+ + n \rightleftarrows p + \bar{\nu}_e, \tag{2.15}$$

$$\nu_e + n \to p + e^-, \qquad \bar{\nu}_e + p \to n + e^+. \tag{2.16}$$

Captures on heavy nuclei and the inverse reactions with thermal neutrinos also contribute in dense, opaque regions, but for the ejecta the reactions on free nucleons dominate.

For matter composed of free nucleons the abundances are $Y_p = Y_e$ and $Y_n = 1 - Y_e$, and the reactions 2.15–2.16 give

$$\frac{dY_e}{dt} = \underbrace{(\lambda_{\nu_e n} + \lambda_{e^+ n})}_{\equiv \lambda_+}(1 - Y_e) - \underbrace{(\lambda_{\bar{\nu}_e p} + \lambda_{e^- p})}_{\equiv \lambda_-} Y_e. \tag{2.17}$$

Where λ_{ab} is the rate of the reaction reactants ab. For slowly varying rates, Y_e relaxes exponentially to the instantaneous equilibrium value

$$Y_e^{\rm eq} = \frac{\lambda_+}{\lambda_+ + \lambda_-} \quad \text{on the timescale} \quad \tau_{\rm weak} = \frac{1}{\lambda_+ + \lambda_-}. \tag{2.18}$$

Whether the ejecta reach equilibrium is decided by the ratio $\tau_{\rm exp}/\tau_{\rm weak}$. If $\tau_{\rm exp} \gg \tau_{\rm weak}$, Y_e follows $Y_e^{\rm eq}$; if $\tau_{\rm exp} \ll \tau_{\rm weak}$, Y_e is frozen. Because both the temperature and the neutrino flux fall quickly as the matter expands, every fluid element eventually freezes out. This is why the slow expanding post-merger ejecta reach higher Y_e than the rapid expanding tidal ejecta which is cold and fast and keeps Y_e similar to the interior of the star before the merger.

2.6 Electromagnetic counterparts

GW170817 showed that a BNS merger is accompanied by emission across the whole electromagnetic spectrum [57]. The gamma-ray burst GRB 170817A was detected 1.7 s after

[1] Here μ refers to chemical potentials

the merger [58]. The optical counterpart AT2017gfo was found 10.9 h later in the galaxy NGC 4993 at $\simeq 40$ Mpc [23]. In addition, an X-ray and radio afterglow appeared days later and kept brightening for several months. At this point we have briefly discussed the formation of the binary, the GW emission, the mass ejection channels and the initial conditions of the ejected material. Now it is time to talk about electromagnetic emissions.

After the merger (seconds) the first emission generated is the gamma ray burst from a relativistic jet, which corresponds to a non thermal radiation. As the density decreases and the ejecta approaches an optically thin regime, thermalized photons escape and build the kilonova (days). Also, some non thermal afterglows are produced when the jet and the ejecta decelerate (hours).

2.6.1 GRBs

Gamma ray bursts (GRBs) are high energy photons released from the polar regions by ultra-relativistic, collimated outflows called jets launched by the remnant. This could be an accreting black hole, which extracts its energy through magnetic fields [59], or a rapidly rotating magnetized neutron star. The photons are produced from the acceleration of the charges inside the jet. When the jet drives into the surrounding medium it decelerates and produces a non thermal afterglow.

However, GRBs are not unique to mergers. Some GRBs are associated with the deaths of massive stars. In the collapsar model the core of the rapidly rotating, stripped massive star collapses to a black hole surrounded by a massive accretion disk, which powers a jet for tens of seconds [60, 61]. The working assumption was therefore that the origin of the GRBs can be identified by the duration. Short GRBs come from mergers and long GRBs from collapsars. But in recent years this was proved wrong. GRB 230307A had a prompt phase of tens of seconds followed by a red kilonova [5, 32], and GRB 200826A had a duration of 0.65s and was followed by a supernova [62]. The duration of a burst is therefore not a reliable indicator of its progenitor. Now the central open question is how a merger can power a jet for tens of seconds, when the accretion time of a typical post-merger disk is $\lessapprox 1$ s.

2.6.2 Kilonovae

Kilonovae belong to the thermal radiation emission since their origin comes from the thermalization of the energy from radioactive decays. However its emission is not exactly the one of a black body, it could have deviations, so a kilonova is a quasi-thermal transient. Most kilonova models, from analytic like LP98 to multidimensional radiative transfer, are built from the same few ingredients: homologous expansion of the ejecta, a radioactive heating rate and its thermalization efficiency, and opacity set by the composition and photon diffusion out of an expanding medium.

Homologous expansion

At early stages the dynamical ejecta is accelerated but shortly after the pressure forces become negligible and every fluid element moves ballistically, $r = vt$. One model for the

density that encapsulates this behavior is the one presented in Ref. [21]

$$\rho(\tau_{\rm exp}, t) = \left\{ \begin{array}{ll} \rho_0 e^{-t/\tau_{\rm exp}} & \text{if } t \leq 3\tau_{\rm exp} \\ \rho_0 \left(\frac{3\tau_{\rm exp}}{et}\right)^3 & \text{if } t \geq 3\tau_{\rm exp} \end{array} \right\}. \tag{2.19}$$

where ρ_0 is an initial density and $\tau_{\rm exp}$ the expansion time. Notice that this model represents a smooth transition from an exponential expansion to a ballistic expansion.

Radioactive heating

As nuclei decay part of their energy stored as mass is released in the form of kinetic energy and photons. Part of the kinetic energy escapes from the system in form of neutrinos and the rest thermalizes through plasma interactions. So effectively not all the energy from decays is used to power the transient. This subsection focuses on the energy and the next subsection will talk about thermalization. Let us first define the nuclear energy released per unit mass from decays, $\dot{\epsilon}$, as

$$\dot{\epsilon}(t) = \sum_{i,x} \frac{Q_{i,x}}{M_{ej}} \lambda_{i,x} N_i(t) = \sum_{i,x} \frac{Q_{i,x}}{m_i} \lambda_{i,x} X_i(t) \tag{2.20}$$

where the index i represents the sum over all the nuclear species and the index x represents the sum over decay modes. $Q_{i,x}$ is the energy released by species i that decay through the decay mode x. $\lambda_{i,x}$ is the decay constant through the decay mode x of species i. X_i is the mass fraction and m_i is the mass of a nucleus i. The alternative version uses M_{ej} (the mass of the ejecta) and N_i the total number of nuclei of species i. At this point this equation does not account for thermalization effects. This definition is completely consistent with the standard definition of heat per baryon due to changes in the composition.

$$\dot{q} = \sum_i (\mu_i + Z_i \mu_e) \dot{Y}_i \tag{2.21}$$

Where Y_i is the number fraction defined as n_i/n which is the number density of i over the number density of baryons, μ_i is the chemical potential of species i. Notice that Eq. 2.20 only accounts for decays while the Eq. 2.21 accounts for any changes in the compositions (decay and reactions). They are consistent in the sense that if we restrict Eq. 2.21 to account for changes only through decays, that means establish an expression for $\dot{Y}_i$, we get Eq. 2.20. Now the issue becomes how to correctly estimate the abundances (X_i,Y_i) as a function of time while taking into account all reactions and decays involved. That problem is solved by implementing a nuclear reaction network which we will treat in detail in Chapter 3. So kilonova models usually implement a reaction network or at least some fits like power laws.

The power-law behavior of $\dot{q}$ can be understood with a statistical argument [3, 63]. Due to the large number of species Eq. 2.20 can be approximated as an integral. From Fermi's theory of allowed β decay the rate scales as, $\lambda \propto Q^5$. Also, $Q \propto \eta$ with $\eta = N - Z$ (neutron excess). Assuming that number of nuclei per neutron excess interval is constant $dN \propto d\eta$[2] it is easy to see that

$$dN = N_0 d\eta = N_0 \lambda^{-4/5} d\lambda, \tag{2.22}$$

[2]The N presented here represents number of nuclei not neutrons

where N_0 is a normalization constant. Let us now calculate the heating rate as follows:

$$\dot{\epsilon} = \sum_i \frac{Q_i \lambda_i}{M_{ej}} N_i(t) \approx \int_0^\infty \frac{1}{M_{ej}} \lambda^{1/5} \lambda N_0 e^{-\lambda t} \lambda^{-4/5} d\lambda \tag{2.23}$$

$$\dot{\epsilon} \propto t^{-7/5} \tag{2.24}$$

in agreement with the analysis of Hotokezaka et al. [63]. Network calculations confirm this behavior, $\dot{q} \simeq 10^{10}\ \mathrm{erg\,g^{-1}\,s^{-1}}\ (t/\mathrm{day})^{-1.3}$ [3, 56]. A widely used fit is [56]

$$\dot{q}(t) = \dot{q}_0 \left[\frac{1}{2} - \frac{1}{\pi} \arctan\left(\frac{t - t_0}{\sigma}\right)\right]^\alpha, \tag{2.25}$$

$$\dot{q}_0 = 2 \times 10^{18}\ \mathrm{erg\,g^{-1}\,s^{-1}},\ t_0 = 1.3\ \mathrm{s},\ \sigma = 0.11\ \mathrm{s},\ \alpha = 1.3, \tag{2.26}$$

which reproduces the plateau during the r-process and the subsequent power law.

Thermalization

Only the fraction of the decay energy that is deposited in the ejecta can be radiated. The effective heating rate is $\dot{\varepsilon}(t) = \sum_x f_\mathrm{x}(t)\, \dot{\epsilon}_x(t)$, with $\dot{\epsilon}_x(t) = \sum_i \dot{\epsilon}_{i,x}(t)$ and f_x the thermalization function of decay mode x. Since thermalization depends on the nature of the decay products then $f_x(t) = \sum_p \chi_p\, f_p(t)$ where χ_p is the average fraction of the energy carried by the product p and p can be γ, α, β or fission daughters. The f_p function is the specific thermalization for the product p. Gamma rays deposit their energy if the ejecta are optically thick to them. A convenient interpolation is $f_\gamma \simeq 1 - \exp(-t_\gamma^2/t^2)$ where t_γ is a function of the mass, velocity and opacity of the ejecta [4, 64]. Most of the gamma-ray energy therefore escapes after about a day, around the time of the peak.

Electrons, α particles and fission fragments are confined by magnetic fields and lose energy by ionization and excitation, at a rate proportional to the density. A convenient way to approximate the thermalization is presented in Ref. [65],

$$f_e(t) \simeq \left(1 + \frac{t}{t_e}\right)^{-n}, \qquad n \simeq 1\text{–}1.5. \tag{2.27}$$

Here t_e is a timescale that depends on the dynamics and mass of the ejecta. This formula can be applied to any charged particle by using different values for n.

Opacity

The free electrons provide only a small opacity, because the ejecta are cool and only weakly ionized. The opacity of kilonova ejecta is instead dominated by the very large number of bound–bound transitions of the r-process elements which are largely unknown. The first models and radiative transfer simulations used relatively low opacities assuming that the optical properties of the ejecta matched the ones from ordinary SNe, mostly related to iron group lines. Such assumptions were later proved wrong as observations matched a kilonova with a high opacity component. Now, one may think that opacity is then related to the number of electrons an atom has but it is rather the large number of distinct ways of moving

electrons within the open shells that increases the opacity. As reference, the number of ways of placing n electrons in a shell with $g = 2(2\ell+1)$ states is $\binom{g}{n}$, at most 252 for a d shell ($g = 10$) but 3432 for an f shell ($g = 14$). The lanthanides and actinides, with open $4f$ and $5f$ shells, therefore have vastly more levels and lines than the iron group, and an opacity higher by one to two orders of magnitude, concentrated at optical and ultraviolet wavelengths [18, 66]. In grey models this is captured by $\kappa \simeq 0.5$–$1\ \mathrm{cm^2\,g^{-1}}$ for lanthanide-poor ejecta ($Y_e \gtrsim 0.3$) and $\kappa \simeq 10$–$30\ \mathrm{cm^2\,g^{-1}}$ for lanthanide-rich ejecta ($X_{\mathrm{La}} \gtrsim 10^{-2}$, $Y_e \lesssim 0.25$)[3].

Diffusion and the time of the peak

Photons produced in the ejecta escape by diffusion. For a uniform sphere the optical depth[4] is $\tau = \kappa\rho R = 3\kappa M/(4\pi v^2 t^2)$, and the diffusion time (the time needed for the photons to reach the surface) is $t_d \simeq \tau R/c$,

$$t_d = \frac{3\kappa M_{ej}}{4\pi\, c\, v\, t} \equiv \frac{\tau_m^2}{t}, \qquad \tau_m \equiv \left(\frac{3\kappa M_{ej}}{4\pi\, c\, v}\right)^{1/2}. \tag{2.28}$$

At early times $t_d \gg t$ and the radiation is trapped and degraded by adiabatic expansion. The emission peaks roughly when $t \simeq \tau_m$.

At this point ends our introductory stage. We have studied what a neutron star is, the different stages of a merger, their mass ejection and we have discussed briefly the GRBs and the kilonovae. The next chapter is going to be fully dedicated to understand the physics involved when implementing a nuclear reaction network. We will cover from nuclear statistical equilibrium to abundance evolution employing decay rates and discuss the thermodynamics involved in the evolution. After that we will retake the discussion on kilonovae and explain in detail the model we will employ in this work. In Chapter 5 we will make use of the theoretical framework built and present how we use it in our research.

[3] X_{La} is defined as the sum of the mass fractions of all lanthanides.

[4] This τ is different from the expansion time τ_{exp} and here we are assuming that the ejecta is already in the ballistic regime (expanding with constant velocity).

Chapter 3

r-Process Nucleosynthesis

3.1 Nuclear statistical equilibrium and freeze-out

3.2 The r-process: neutron-to-seed ratio, fission cycling

3.3 Reaction-network equations and numerical solution

3.4 Nuclear inputs: masses, decay rates, neutron capture, fission

3.5 Radioactive heating and thermalization

Chapter 4

Kilonova Light Curves

4.1 Li-Paczyński model

4.2 Radiative transfer and the diffusion regime

4.3 Opacities of r-process matter

4.4 Light-curve modeling

Chapter 5

Methodology on the impact of theoretical decay rates across thermodynamics trajectories

5.1 Binary neutron-star simulation

5.2 From simulation to trajectories: ejecta extraction and expansion

5.3 Initial termodinamic trajectories treatment

5.4 Nuclear reaction network setup and theoretical reaction reates

5.5 Modified Li-Paczyński model

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 6

Results

6.1 Nucleosynthesis and heating rates

6.1.1 Relevant α and β

6.2 Kilonova light curves

6.3 Detectability with current and future telescopes

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 7

Discussion

7.1 Comparison with previous studies

7.2 Limitations and model assumptions

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 8

Conclusions and Outlook

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Appendix A

Quadrupole and L_{GW}

In this appendix we will go through the details of finding the mass quadrupole components, explain the approximations assumed and finding the L_{GW}. Lets start with the definitions, as we saw from Fig. 2.2 the position are:

$$\mathbf{r}_1 = -\frac{\mu a}{m_1}(cos(\theta), sin(\theta), 0), \tag{A.1}$$

$$\mathbf{r}_2 = \frac{\mu a}{m_2}(cos(\theta), sin(\theta), 0). \tag{A.2}$$

$$\tag{A.3}$$

We used $||\mathbf{r}_1|| = m_2 a/(m_1 + m_2) = \mu a/m_1$ with $\mu = m_1 m_2/(m_1 + m_2)$. The first quadrupole component will be:

$$Q_{xx} = m_1 r_{x1} r_{x1} + m_2 r_{x2} r_{x2} = m_1 \frac{\mu^2 a^2}{m_1^2} cos^2(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} cos^2(\theta), \tag{A.4}$$

$$Q_{xx} = \mu^2 a^2 cos^2(\theta)(\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.5}$$

$$Q_{xx} = \mu a^2 cos^2(\theta), \tag{A.6}$$

$$Q_{xx} = \frac{\mu a^2}{2}(cos(2\theta) + 1). \tag{A.7}$$

$$\tag{A.8}$$

Similarly for Q_{yy}:

$$Q_{yy} = m_1 r_{y1} r_{y1} + m_2 r_{y2} r_{y2} = m_1 \frac{\mu^2 a^2}{m_1^2} sin^2(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} sin^2(\theta), \tag{A.9}$$

$$Q_{yy} = \mu^2 a^2 sin^2(\theta)(\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.10}$$

$$Q_{yy} = \mu a^2 sin^2(\theta), \tag{A.11}$$

$$Q_{yy} = \frac{\mu a^2}{2}(-cos(2\theta) + 1). \tag{A.12}$$

$$\tag{A.13}$$

Finally for Q_{xy}

$$Q_{xy} = m_1 r_{x1} r_{y1} + m_2 r_{x2} r_{y2} = m_1 \frac{\mu^2 a^2}{m_1^2} sin(\theta) cos(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} sin(\theta) cos(\theta), \tag{A.14}$$

$$Q_{xy} = \mu^2 a^2 sin(\theta) cos(\theta) (\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.15}$$

$$Q_{xy} = \mu a^2 sin(\theta) cos(\theta), \tag{A.16}$$

$$Q_{xy} = \frac{\mu a^2}{2} sin(2\theta). \tag{A.17}$$

$$\tag{A.18}$$

Now we need the third derivate with respect to time. We will assume two things, $\ddot{\theta} << 1$ and $\dot{a} << 1$. Effectively we are saying that the change in angular velocity and the change in the radius is slow which is referred as an adiabatic process. To see how good the approximation is we can compare the orbit time scale with the inspiral timescale. Which in fact is the situation, therefore is a good approximation until the merger itself. Therefore it is easy to see that:

$$\dddot{Q}_{xx} = 4\mu a^2 \dot{\theta}^3 sin(2\theta) \tag{A.19}$$

$$\dddot{Q}_{yy} = -4\mu a^2 \dot{\theta}^3 sin(2\theta) \tag{A.20}$$

$$\dddot{Q}_{xy} = -4\mu a^2 \dot{\theta}^3 cos(2\theta) \tag{A.21}$$

$$\tag{A.22}$$

Now we meed to calculate the following term remembering that we are using einstein notation for the repeated indices and that $Q_{xy} = Q_{yx}$:

$$\dddot{Q}_{ij} \dddot{Q}_{ij} - \dddot{Q}_{ii} \dddot{Q}_{jj} = \dddot{Q}_{xx} \dddot{Q}_{xx} + \dddot{Q}_{xy} \dddot{Q}_{xy} + \dddot{Q}_{yx} \dddot{Q}_{yx} + \dddot{Q}_{yy} \dddot{Q}_{yy} \tag{A.23}$$

$$- \dddot{Q}_{xx} \dddot{Q}_{xx} - \dddot{Q}_{yy} \dddot{Q}_{xx} - \dddot{Q}_{xx} \dddot{Q}_{yy} - \dddot{Q}_{yy} \dddot{Q}_{yy} \tag{A.24}$$

$$= \dddot{Q}_{xx}^2 + 2\dddot{Q}_{xy}^2 + \dddot{Q}_{yy}^2 - \dddot{Q}_{xx}^2 - 2\dddot{Q}_{yy} \dddot{Q}_{xx} - \dddot{Q}_{yy}^2 \tag{A.25}$$

$$= \dddot{Q}_{xx}^2 + 2\dddot{Q}_{xy}^2 + \dddot{Q}_{yy}^2 - (\dddot{Q}_{xx} + \dddot{Q}_{yy})^2 \tag{A.26}$$

$$= 16\mu^2 a^4 \dot{\theta}^6 (sin^2(2\theta) + 2cos^2(2\theta) + sin^2(2\theta)) \tag{A.27}$$

$$= 32\mu^2 a^4 \dot{\theta}^6 = \frac{32 G^3 m_1^2 m_2^2 M}{a^5} \tag{A.28}$$

Finally:

$$L_{GW} = \frac{G}{5c^5} (\dddot{Q}_{ij} \dddot{Q}_{ij} - \dddot{Q}_{ii} \dddot{Q}_{jj}) = \frac{32 G^4 m_1^2 m_2^2 M}{c^5 a^5} \tag{A.29}$$

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